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Studierichting: Informatica
Oefeningenreeks 3B nr. 2.3

$$f(x) = (x^2 - 1)^4$$

$$f'(x) = 4(x^2 - 1)^3 \cdot \frac{d}{dx}(x^2 - 1)$$

$$= 4(x^2 - 1)^3 \cdot 2x$$

$$= 8x(x^2 - 1)^3$$

$$f''(x) = \frac{d}{dx}(8x) \cdot (x^2 - 1)^3 + 8x \cdot \frac{d}{dx}\left((x^2 - 1)^3\right)$$

$$= 8(x^2 - 1)^3 + 8x \cdot 3(x^2 - 1)^2 \cdot 2x$$

$$= 8(x^2 - 1)^3 + 48x^2(x^2 - 1)^2$$

$$= 8(x^6 - 3x^4 + 3x^2 - 1) + 48x^2(x^4 - 2x^2 + 1)$$

$$= 56x^6 - 120x^4 + 72x^2 - 8$$

$$f'''(x) = \frac{d}{dx}(56x^6) - \frac{d}{dx}(120x^4) + \frac{d}{dx}(72x^2) - \frac{d}{dx}(8)$$

$$= 336x^5 - 480x^3 + 144x$$

References